The Nix Series: A Proposed Combinatorial Spectral Expansion for Nonlinear Frequency Interactions

Nishanth-inspired working paper: formalization, synthetic experiments, and candidate application domains

Abstract

Linear spectral methods such as the Fourier transform report which frequencies are present and their net amplitudes, but they do not natively describe how nonlinear interactions between active frequencies generate new spectral components or redistribute amplitude among existing lanes. This paper proposes the *Nix series*, a combinatorial spectral expansion for weakly to moderately nonlinear frequency-domain systems. The core idea is to treat nonlinear spectral behavior as a graded operator acting on a base frequency set and generating an expanded closure of combinatorial frequencies. We formalize the closure operator, define truncated Nix expansions and a practical sparsity rule, introduce a forward *Nix transform* and an inverse approximation problem, and give a weakly nonlinear resonant transducer model inspired by piezoelectric wave-based compute prototypes. We then run purely synthetic experiments to test internal consistency and practical usefulness. The experiments show four things: (i) the Nix closure grows rapidly but predictably with order, (ii) sparse Nix features outperform a linear baseline and match dense degree-3 polynomial models with far fewer features, (iii) a greedy Nix-aware bin-selection strategy is near-optimal in a simulated spectral planning task, and (iv) operating inside a lane-wise linear regime via scaling or chunked accumulation greatly improves synthetic analog dot-product accuracy. The paper is positioned as a theory-and-simulation primer, not an empirical validation of any specific device. Its intended contribution is a mathematically explicit language for nonlinear spectral computation and a roadmap for future empirical work in analog compute, communications, signal processing, and machine learning.

1. Introduction

Spectral representations are among the most successful abstractions in mathematics, signal processing, and scientific computing. The fast Fourier transform (FFT) turned Fourier analysis into a practical algorithmic primitive and changed entire engineering disciplines [**?**]. Yet Fourier analysis is fundamentally linear: it reports the spectrum of a signal with respect to a fixed basis, but it does not by itself encode how nonlinear interactions among active frequencies create new components such as $f_i + f_j$, $|f_i - f_j|$, or $2f_i - f_j$.

In real physical systems, these combinatorial frequencies appear everywhere: power amplifiers generate intermodulation products, piezoelectric and acoustic structures exhibit mode coupling, and nonlinear optical media generate sum- and difference-frequency components. The classical Volterra framework offers a powerful general language for nonlinear systems [**?**, **?**]; however, in practice its kernel growth is severe and its time-domain formulation is not always the most natural language for frequency-coded computation.

This paper proposes a complementary viewpoint: treat nonlinear spectral behavior itself as the primitive. The proposed *Nix series* is a graded combinatorial spectral expansion in which amplitudes on a base frequency set induce amplitudes on an expanded closure of combinatorially generated frequencies. The proposal is motivated by weakly nonlinear multi-tone analog compute settings, but the formalization is intentionally more general.

The goals of this paper are modest and explicit:

- formalize the Nix closure and the truncated Nix series;
- define a practical sparsity rule for weakly nonlinear resonant systems;

- define a forward Nix transform and an inverse approximation problem;
- instantiate the framework with a PZT-inspired resonant transducer model;
- test the framework using algorithmically generated synthetic experiments.

The paper does *not* claim empirical validation of a new law of nature. Instead, it offers a mathematically explicit proposal that can be falsified, extended, or refined by future experiments.

2. Formal definitions

2.1. Base frequency set and closure

Let the base frequency set be

$$F^{(0)} = \{f_1, f_2, \ldots, f_n\}, \qquad f_i > 0. \tag{1}$$

Let the corresponding amplitude vector be $\boldsymbol{a} = (a_1, \ldots, a_n)^\top$, with $a_i \in \mathbb{R}$ or $\mathbb{C}$.

Definition 1 (Nix closure operator)**.** *For truncation order $p \in \mathbb{N}$, define the* Nix closure

$$\mathcal{F}_p(F^{(0)}) = \left\{ \left| \sum_{i=1}^{n} m_i f_i \right| \,\middle|\, m_i \in \mathbb{Z},\ \sum_{i=1}^{n} |m_i| \le p,\ \sum_{i=1}^{n} |m_i| \ge 1 \right\}. \tag{2}$$

The set contains all frequencies obtainable by signed integer combinations of the base frequencies up to total interaction order p.

This closure is the first point of departure from ordinary multilinear tensor maps: the output index set is not fixed in advance but *generated* by combinatorial interaction.

Proposition 1 (Collision-free closure count upper bound)**.** *If distinct coefficient vectors $\boldsymbol{m} \in \mathbb{Z}^n$ with $\|\boldsymbol{m}\|_1 \le p$ generate distinct positive frequencies, then*

$$|\mathcal{F}_p(F^{(0)})| \le \sum_{s=1}^{p} \sum_{k=1}^{\min(n,s)} \binom{n}{k} 2^k \binom{s-1}{k-1}. \tag{3}$$

Proof. For fixed total order s, choose a support of size k, assign signs to the k nonzero coefficients, and distribute total absolute weight s across the selected coordinates. The number of positive integer compositions of s into k parts is $\binom{s-1}{k-1}$. Summing over admissible k and then over $s = 1, \ldots, p$ yields the bound. □

The bound is achieved when the base frequencies are sufficiently noncommensurate that collisions are rare.

2.2. The truncated Nix series

Definition 2 (Truncated Nix series)**.** *Let $\Omega_p = \mathcal{F}_p(F^{(0)})$ be the closure up to order p. A* truncated Nix series of order p *is a mapping*

$$\mathcal{N}_p : \mathbb{R}^n \to \mathbb{R}^{|\Omega_p|} \tag{4}$$

of the form

$$A(\omega) = \sum_{r=1}^{p} \sum_{\alpha \in \mathcal{I}_r(\omega)} C_\alpha^{(\omega)} \prod_{q=1}^{r} a_{i_q}, \qquad \omega \in \Omega_p, \tag{5}$$

where $\mathcal{I}_r(\omega)$ is the set of order-r interaction tuples whose combinatorial rule generates frequency ω, and $C_\alpha^{(\omega)}$ are interaction coefficients.

The representation is naturally *graded* by interaction order:

$$\boldsymbol{A} = \mathbf{C}^{(1)}\boldsymbol{a} + \mathbf{C}^{(2)}(\boldsymbol{a} \otimes \boldsymbol{a}) + \mathbf{C}^{(3)}(\boldsymbol{a} \otimes \boldsymbol{a} \otimes \boldsymbol{a}) + \cdots + \mathbf{C}^{(p)}(\boldsymbol{a}^{\otimes p}). \tag{6}$$

The novelty is not the existence of higher-order tensors per se; it is that the output is indexed on the *expanded combinatorial closure* rather than solely on the original base set.

2.3. A practical sparsity rule

In physical systems, the full closure is usually far larger than the practically significant set of frequencies. We therefore propose a weakly nonlinear sparsity rule.

Assumption 1 (Weakly nonlinear resonant sparsity)**.** *In many practical systems, the significant Nix coefficients satisfy three structural biases:*

1. ***Low-order dominance:*** *second- and third-order coefficients dominate fourth order and beyond.*
2. ***Near-resonant support:*** *only combinatorial products close to resonant or observed lanes retain appreciable amplitude.*
3. ***Local interaction structure:*** *coefficient magnitude decays with spectral separation, mode mismatch, or both.*

This converts the Nix series from a combinatorial explosion into a tractable sparse model. In practice, this sparsity is what makes the framework useful rather than merely descriptive.

2.4. Nix transform and inverse approximation

We separate two computational problems.

Forward Nix transform. Given amplitudes $\boldsymbol{a}$ on the base set and coefficient tensors up to order p, the forward problem is simply evaluation of $\mathcal{N}_p(\boldsymbol{a})$.

Inverse approximation. Given a measured closure-spectrum $\boldsymbol{y} \in \mathbb{R}^{|\Omega_p|}$, recover a base amplitude vector $\boldsymbol{a}$ that approximately explains it:

$$\hat{\boldsymbol{a}} = \arg\min_{\boldsymbol{a}\in\mathbb{R}^n} \|\mathcal{N}_p(\boldsymbol{a}) - \boldsymbol{y}\|_2^2 + \lambda\|\boldsymbol{a}\|_2^2. \tag{7}$$

For weakly nonlinear settings, a Gauss–Newton or Levenberg–Marquardt scheme is natural; for sparse spectral systems, one may additionally regularize toward sparse supports or bounded amplitudes.

2.5. A PZT-instantiated weakly nonlinear model

To anchor the abstractions, consider a weakly nonlinear resonant transducer array or split-terminal piezoelectric structure with base bins f_i. A useful low-order surrogate is

$$A(\omega) = \sum_i g_i^{(\omega)} a_i + \sum_{i,j} q_{ij}^{(\omega)} \eta_{ij} a_i a_j + \sum_{i,j,k} r_{ijk}^{(\omega)} \eta_{ijk} a_i a_j a_k, \tag{8}$$

where:

- $g_i^{(\omega)}$ are lane-wise linear gains,

- $q_{ij}^{(\omega)}$ model second-order interactions,
- $r_{ijk}^{(\omega)}$ model third-order interactions,
- η_{ij} and η_{ijk} encode geometry, boundary conditions, terminal asymmetry, or mode-overlap penalties.

This model is not offered as a universal piezoelectric law; it is a practical weakly nonlinear instantiation of the Nix framework.

3. Synthetic experimental program

All experiments in this paper are synthetic. The purpose is not to establish a physical law but to ask whether the Nix formalism is computationally coherent and whether it yields useful algorithmic behavior under controlled nonlinear scenarios.

3.1. Experiment 1: closure growth

We compute the cardinality of $\mathcal{F}_p(F^{(0)})$ for $n \in \{3, 4, 5\}$ base bins and truncation orders $p \in \{1, \ldots, 5\}$ using noncommensurate synthetic frequencies. This tests whether the combinatorial closure grows in the way the counting proposition predicts.

3.2. Experiment 2: prediction on synthetic nonlinear spectra

We generate a synthetic weakly nonlinear system with $n = 5$ base bins, sparse second-order support on selected pairs, and sparse third-order support on selected triples. We compare three regression models on held-out data:

1. a linear-only baseline,
2. a dense degree-3 polynomial model,
3. a sparse order-3 Nix feature set defined only on the active closure interactions.

The question is whether a sparse Nix parameterization can match the dense cubic model with many fewer features while decisively outperforming a purely linear fit.

3.3. Experiment 3: greedy Nix-aware bin selection

We simulate 12 candidate frequency bins, each with a single-tone utility score and a pairwise interference penalty derived from spacing, harmonic coincidence, and accidental sum/difference collisions. We compare:

1. top-gain-only selection,
2. a greedy Nix-aware selector maximizing gain minus interaction cost,
3. brute-force optimal search.

This tests whether a Nix-aware planning rule is useful for practical multi-tone bin selection.

3.4. Experiment 4: linear-regime scheduling

We simulate a three-lane analog dot-product channel with amplitude-dependent distortion and pairwise cross-lane leakage. We compare three execution policies over random positive vectors:

1. one unscaled analog pass,
2. global scaling with digital rescaling,
3. chunked accumulation, where large amplitudes are decomposed into multiple safe passes.

The purpose is to test a direct design implication of the Nix viewpoint: if nonlinear contamination is structured, then operating inside a lane-wise linear regime should sharply reduce error.

4. Results

4.1. Closure growth is rapid but predictable

Figure **??** shows closure growth for 3, 4, and 5 base bins. The growth is fast, but it remains systematically order-graded rather than arbitrary. For five base bins, the closure grew from 5 frequencies at order 1 to 841 by order 5 in the collision-light synthetic setting.

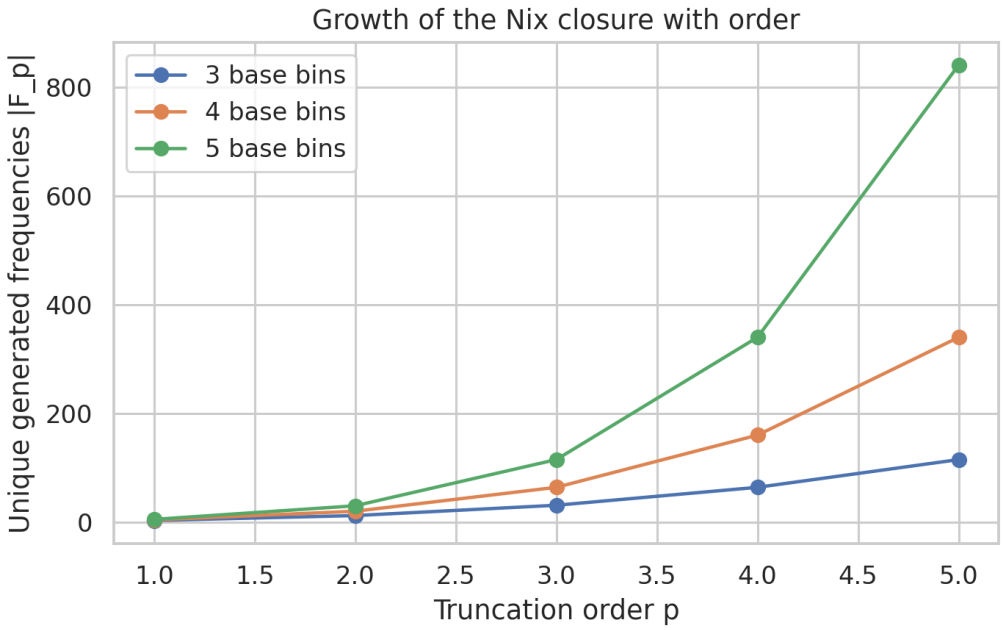


Figure 1: Growth of the Nix closure with truncation order for synthetic noncommensurate base frequencies.

4.2. Sparse Nix features recover nonlinear spectra efficiently

Figure **??** compares test RMSE as the synthetic nonlinearity strength α increases. The linear baseline degrades monotonically, while the sparse Nix model matches the dense cubic model with many fewer features (16 versus 55 in this experiment). At $\alpha = 1.0$, the linear baseline had RMSE

0.068, whereas the sparse Nix model achieved 0.0103, essentially matching the dense cubic fit at 0.0112.

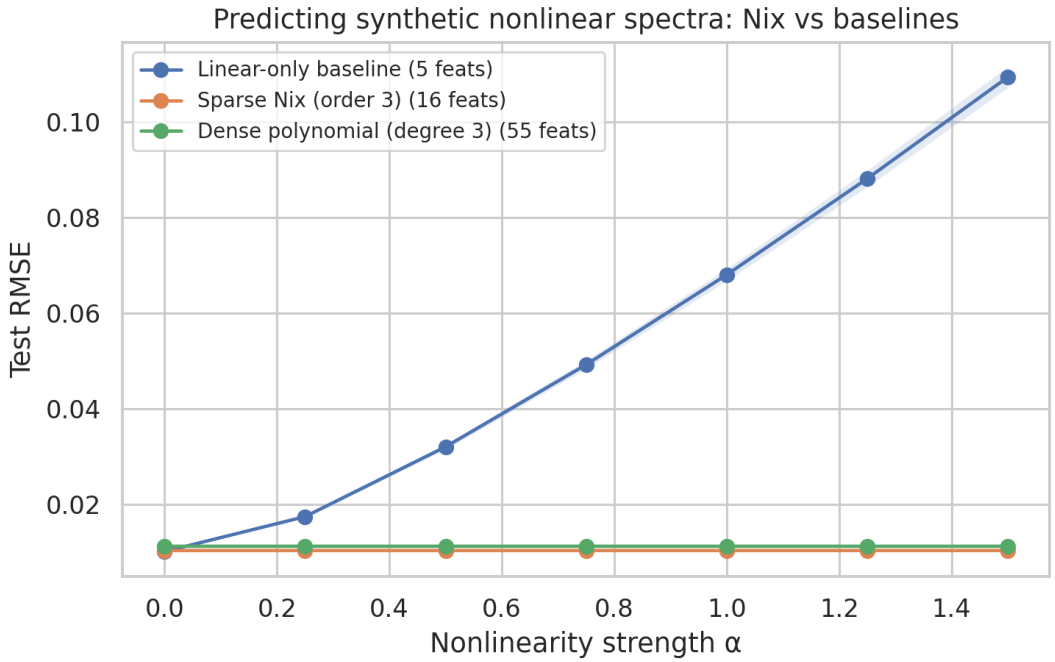


Figure 2: Prediction accuracy on synthetic nonlinear spectra. A sparse order-3 Nix model matches a dense cubic model with substantially fewer features and strongly outperforms a linear-only baseline as nonlinearity grows.

4.3. Greedy Nix-aware bin selection is near-optimal

Figure **??** shows the median selection objective normalized by the brute-force optimum across 300 synthetic planning problems. The greedy Nix-aware selector reached the optimum or near-optimum almost always (mean 0.992 of optimum), whereas top-gain-only selection achieved only 0.781 of the median optimum and incurred substantially higher pairwise interference penalties.

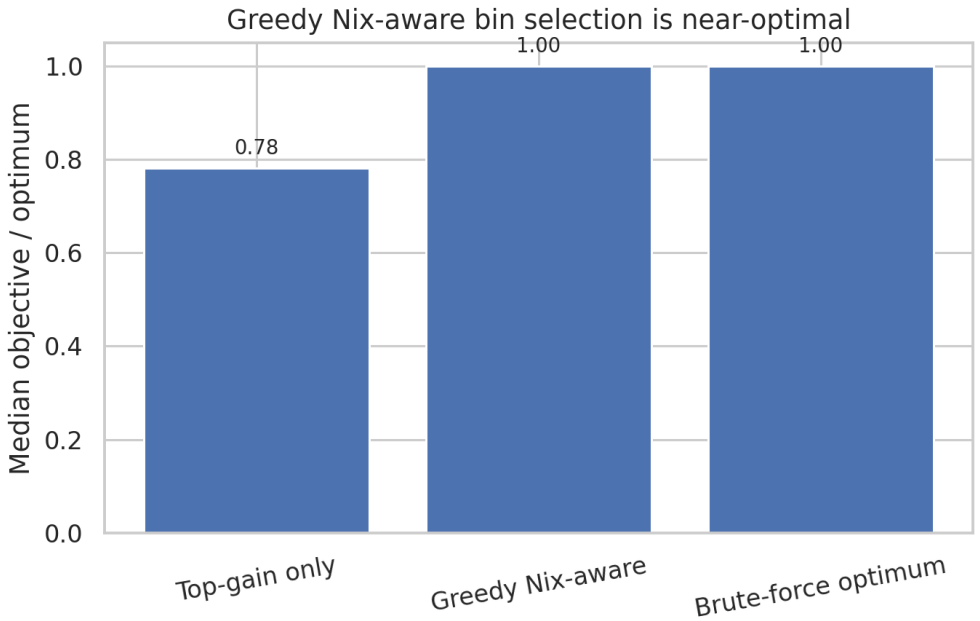


Figure 3: Median objective ratio for frequency-bin selection strategies. Greedy Nix-aware selection is near-optimal under the synthetic interaction model.

4.4. Operating inside the linear regime matters

Figure **??** shows median relative error for the synthetic analog dot-product experiment across dynamic-range quintiles. The unscaled one-pass strategy exhibited high error (median around 0.59 overall), while global scaling plus rescaling reduced median relative error to 0.0097 and chunked accumulation reduced it further to 0.0066.

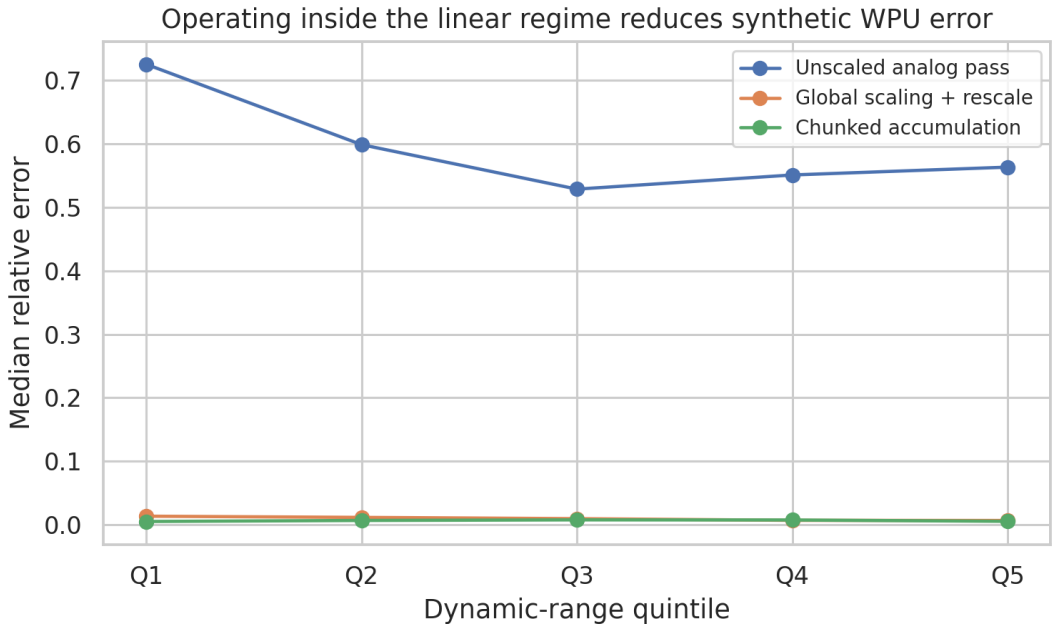


Figure 4: Synthetic analog compute experiment: scheduling the computation to remain within a safe regime dramatically reduces error.

4.5. Condensed numerical summary

Table **??** collects representative synthetic metrics.

Table 1: Condensed numerical results from the synthetic program.

Metric	Linear / naive	Nix-aware / structured	Notes
Prediction RMSE at $\alpha = 1.0$	0.0680	0.0103	sparse Nix (16 features) vs linear baseline (5 features)
Prediction RMSE at $\alpha = 1.5$	0.1094	0.0103	dense cubic: 0.0112 with 55 features
Median objective / optimum in bin selection	0.781	1.000	greedy Nix-aware median matched optimum to 3 decimals
Mean objective / optimum in bin selection	0.111	0.992	top-gain mean hurt by severe interference outliers
Median relative error, analog dot product	0.5851	0.0097 / 0.0066	global scaling / chunked accumulation

5. Applicable domains

The Nix formalism is intentionally general. The following domains appear most natural.

5.1. Analog compute

This is the most immediate domain. Multi-tone analog compute systems naturally require a language for bin planning, cross-lane leakage modeling, calibration, and scheduling within lane-wise linear regimes. The synthetic experiments suggest that a Nix-aware control layer can be useful even before a full device law is known.

5.2. RF and wireless communications

Communications systems routinely confront intermodulation, sum/difference contamination, and adjacent-channel effects. A Nix-style sparse closure model could be useful for nonlinear equalization, carrier allocation, amplifier-aware scheduling, and side-product cancellation.

5.3. Acoustics and ultrasonics

Piezoelectric, surface-acoustic-wave, and ultrasonic systems already exhibit rich nonlinear spectral behavior. A closure-based combinatorial model gives a natural language for mode interaction and harmonic planning.

5.4. Photonics and nonlinear optics

Sum-frequency and difference-frequency generation are already standard physical processes in optics. The value of Nix here would be to provide a computationally explicit sparse algebra rather than only a device-specific physical derivation.

5.5. Signal processing

Linear transforms changed signal processing because they revealed structure that could be exploited algorithmically. Nix does not replace Fourier analysis; it instead proposes a possible language for *nonlinear spectral structure*. Candidate future directions include Nix-domain denoising, sparse nonlinear filtering, and combinatorial spectral compression.

5.6. Machine learning

If nonlinear spectral interaction patterns can be learned or regularized efficiently, one can imagine Nix-inspired layers for harmonic-rich signals, scientific machine learning, and analog-aware feature extractors. At present this remains speculative, but the algebraic pathway is clear.

6. Discussion

Three features make the proposal potentially useful.

First, the Nix closure is explicit. Instead of speaking vaguely about "distortion" or "cross-frequency leakage," the model says that nonlinear systems generate an expanded spectral basis according to combinatorial rules.

Second, the proposal is sparse by design. Without the sparsity rule, the closure becomes too large for practical use. With it, the framework behaves like a tractable order-graded algebra rather than an intractable combinatorial explosion.

Third, the synthetic experiments already suggest concrete engineering payoffs: closure-aware regression, closure-aware bin selection, and scheduling policies that keep analog computation inside a safe operating regime.

The most important limitations are equally clear. The present paper does not prove a universal physical law, does not identify a unique inverse transform, and does not derive coefficients from first principles for any specific device. Those tasks belong to future empirical work.

7. Research roadmap

A concrete next-step program is as follows:

1. **Empirical coefficient estimation:** estimate low-order Nix coefficients on a controlled physical platform and test reproducibility under repeated mounting and drive conditions.
2. **Closure pruning laws:** determine whether coefficient magnitude follows a simple empirical decay law in spectral separation, resonance mismatch, or interaction order.
3. **Inverse Nix algorithms:** develop stable approximate inversion methods, ideally exploiting sparsity and order truncation.
4. **Hardware co-design:** use Nix-aware bin selection and amplitude scheduling to design analog compute operating regions.
5. **Algorithmic export:** test whether Nix-inspired sparse nonlinear spectral models improve communications, DSP, or ML tasks beyond the originating hardware context.

8. Conclusion

This paper introduced the Nix series as a proposed combinatorial spectral expansion for nonlinear frequency interactions. The key conceptual move is to allow the spectral basis to expand under combinatorial interaction rather than remain fixed, and to model that expansion with sparse order-wise tensors defined on the resulting closure. Synthetic experiments show that the framework is internally coherent and algorithmically useful: sparse Nix features recover nonlinear spectra efficiently, Nix-aware bin planning is near-optimal in a simulated setting, and operating inside a lane-wise linear regime dramatically improves synthetic analog compute accuracy. The proposal is therefore best viewed as a new mathematical and computational language candidate for weakly nonlinear spectral systems.

Reproducibility statement

All experiments reported here are synthetic and were generated algorithmically for this paper. They test internal coherence of the formalism, not empirical truth of a physical device law.

References

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